



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

T-SQL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is T-SQL and what are its core strengths? [Threat Model](#)

Q2 How does T-SQL guarantee consistency and handle transactions? [Architecture](#)

Q3 What index types does T-SQL support and when do you use each? [Indexing](#)

Q4 How do you design a schema or data model in T-SQL? [Hardening](#)

Q5 How does T-SQL scale horizontally? [SSO / IdP](#)

Q6 How do you monitor and tune slow queries in T-SQL? [Observability](#)

Q7 What backup and restore strategies does T-SQL support? [Lifecycle](#)

Q8 How do you handle schema migrations in T-SQL? [Compliance](#)

Q9 What security features does T-SQL provide? [Testing](#)

Q10 What are common anti-patterns when using T-SQL in production? [Tuning](#)



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 T-SQL is a relational, document, key-value, graph,

Mitigation

T-SQL is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 T-SQL provides a slow query log, explain/explain

Observability

T-SQL provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 T-SQL provides either full ACID transactions or

Primitives

T-SQL provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 T-SQL supports logical backups (export SQL or

Automation

T-SQL supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 T-SQL offers B-tree, hash, full-text, spatial, or

Hardening

T-SQL offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in T-SQL are managed with

Governance

Schema migrations in T-SQL are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in T-SQL involves normalizing relations

Gotchas

Schema design in T-SQL involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 T-SQL supports role-based access control, encrypted

Assessment

T-SQL supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 T-SQL scales via replication (read replicas) for

Federation

T-SQL scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some T-SQL implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all columns instead

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.